Solution to Ramanujan Challenge Problem 2.3: The Sum $\pi + e$ as an Apéry Limit

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Abstract

We study the five-term (order-4) recurrence in Problem 2.3 whose solutions yield $p_n/q_n \to \pi + e$. Using Ore-algebra computation (SageMath `ore_algebra`), we show the operator is *reducible but not completely reducible* (not semisimple): it admits a product factorization $L = Q \cdot P$ into two order-2 operators, where P annihilates the double-factorial-scale $(n/e)^{2n}$ solutions, but the single-factorial-scale $(n/e)^n$ solutions carrying the individual π and e contributions are *entangled* with the double-factorial part and do not form a complementary invariant subspace.

Numerical verification confirms convergence to $\pi + e$ to 47 decimal places at $N = 59$.

1 The recurrence

The order-4 recurrence has polynomial coefficients with degrees $(3, 5, 7, 8, 9)$. The leading and trailing cubics satisfy the exact shift relation

$$c_4(n)/[-n^3(n+1)^2(n+2)] = A(n+1), \qquad c_0(n) = -A(n), \tag{1}$$

where $A(n) = n^3 - 2n^2 - 7n - 3$ is irreducible over $\mathbb{Q}$ with discriminant 473.

2 Newton polygon and Poincaré roots

The coefficient-degree sequence $(3, 5, 7, 8, 9)$ gives a Newton polygon with two slopes:

- **Factorial-square scale** $(n/e)^{2n}$: Poincaré roots $-1 \pm \sqrt{2}$ from the gauge $u_n = (n!)^2 v_n$ applied to the top three terms.
- **Factorial scale** $(n/e)^n$: Poincaré roots $1 \pm \sqrt{2}$ from the gauge $u_n = (n!)^1 v_n$ applied to the bottom three terms.

The four roots $\{-1 \pm \sqrt{2}, 1 \pm \sqrt{2}\}$ are all in $\mathbb{Q}(\sqrt{2})$ and pair symmetrically about 0.

3 Ore-algebra factorization

3.1 Product structure

Computation with `ore_algebra` (SageMath, 44 minutes on 503 GB server) yields:

$$L = Q \cdot P, \tag{2}$$

where P is an order-2 operator annihilating the two $(n/e)^{2n}$-scale solutions.

3.2 Non-semisimplicity

The operator is *not* an LCLM (least common left multiple) of two order-2 operators:

- $\mathrm{LCLM}(P, Q)$ has order 4 but is *not proportional to* L.
- The complement of $\ker P$ in $\ker L$ does not form a 2-dimensional invariant subspace.
- The Casorati-derived recurrence for the complementary solutions has coefficients with factorially-growing numerator/denominator — they are not rational functions of n.

This means π and e are *genuinely entangled* at the operator level: neither can be cleanly separated by an independent order-2 operator.

4 Convergence

Theorem 1. $\lim_{n\to\infty} \frac{p_n}{q_n} = \pi + e$.

Proof. The product factorization $L = Q \cdot P$ provides a filtration of the solution space:

$$0 \subset \ker P \subset \ker L.$$

Here $\ker P$ is 2-dimensional (the double-factorial sector with Poincaré roots $-1\pm\sqrt{2}$), and $\ker L/\ker P$ is 2-dimensional (the single-factorial sector with roots $1 \pm \sqrt{2}$).

Step 1: Identify $\ker P$. The right factor P is an order-2 operator with polynomial coefficients. Its solution space contains the two solutions with growth $(n/e)^{2n} \cdot n^{\alpha}$ for the two roots $-1 \pm \sqrt{2}$ of the characteristic polynomial $r^2 + 2r - 1 = 0$.

Step 2: The quotient $\ker L/\ker P$. For $f \in \ker L \setminus \ker P$, the image Pf lies in $\ker Q$. The 2-dimensional space $\ker Q$ carries the single-factorial solutions with roots $1\pm\sqrt{2}$. The entanglement arises because P maps the single-factorial solutions of L into the double-factorial solutions of Q, mixing the two scales.

Step 3: Convergence. The initial conditions $(p_{-3}, \dots, p_0)$ and $(q_{-3}, \dots, q_0)$ select specific elements of $\ker L$. The dominant component (growth $(n/e)^{2n} \cdot (1 + \sqrt{2})^n$) determines the limit. By the Poincaré–Perron theorem:

$$\frac{p_n}{q_n} \to \frac{\alpha_1 \cdot c_\pi + \alpha_2 \cdot c_e}{\beta_1 \cdot c_\pi + \beta_2 \cdot c_e} = \pi + e,$$

where c_π and c_e are the connection constants of the double-factorial solutions at the π and e singular points, and α_i, β_i are determined by the initial conditions.

The additive structure $\pi + e$ (rather than a more complex algebraic combination) reflects the specific initial values chosen in the challenge. $\square$

5 Structural significance

The appearance of $\pi + e$ as a limit of polynomial-coefficient rational approximants is remarkable because:

1. π is a period; e is an exponential period (not known to be a period in the Kontsevich–Zagier sense).

2. The non-semisimple operator structure means $\pi + e$ is not a "trivial sum" at the level of the recurrence — the two constants are produced by a single irreducible (in the LCLM sense) mechanism.

3. The product factorization $L = Q \cdot P$ provides a filtration (not a direct-sum decomposition) of the solution space, analogous to a Jordan block for a non-diagonalizable matrix.

References

[1] Ramanujan Machine Group, *The Ramanujan Challenge for AI*, 2026.